

SYARFANDI "ANDI" ACHMAD

Data Engineer Makassar, Indonesia | [+62 852-5652-2335](tel:+62852-5652-2335) | syarfandi.achmad@gmail.com | [Portfolio: syarfandi.github.io](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Strategic Data Engineer with 4+ years of experience in architecting scalable data pipelines and robust database environments. Expert in ETL/ELT processes, high-performance PostgreSQL tuning, and real-time data ingestion using Supabase and Kafka. Proven success in consolidating regional-scale data (9M+ records) and managing complex retail data layers. Committed to delivering data integrity and high-availability systems for mission-critical analytics.

TECHNICAL SKILLS

- **Programming & Scripting:** Python (Airflow, Pandas, PySpark), SQL (Advanced), Bash
- **Data Engineering:** ETL/ELT Pipeline Design, Data Modeling, Schema Design, Data Migration
- **Databases:** PostgreSQL (High-Availability, Tuning, Replication), Supabase, MySQL/MariaDB
- **Data Warehousing:** Google BigQuery, Snowflake, Amazon Redshift
- **Real-time Data:** Kafka, RabbitMQ, Webhooks, Supabase Realtime
- **Infrastructure & Cloud:** GCP, AWS, Terraform, Docker, Kubernetes, Proxmox VE
- **Monitoring & BI:** Zabbix, Prometheus, Grafana, Tableau, Google Looker Studio

FEATURED PROJECTS

- [Kawan Sedarah \(Blood Stock\)](#): A **command center** dashboard that tracks **blood stock** availability across donor units in **real-time**, powered by **Supabase** as its core backend.
- [Portal Data Sulsei](#): A unified information system delivering **data visualization**, **monitoring**, and religious **statistical analysis** across all districts in South Sulawesi in **real-time**.
- [AI Virtual Assistant](#): A **Conversational AI**-powered virtual assistant featuring **Realtime Lipsync** on 3D avatars, backed by **Voice AI** and **LLM** for immersive conversational interactions.
- [GetKasir Enterprise POS](#): A large-scale **Enterprise** POS ecosystem with **100+ modules**, integrating **AI-driven** sales prediction, **biometric** authentication, and a scalable **cloud-native** architecture.
- [PPID Kemenag Kota Parepare](#): A public information disclosure portal for **PPID Kemenag** providing access to **official documents**, **performance reports**, and an intelligent chatbot powered by **Google Gemini AI**.
- [OP Tracker](#): A premium logbook for tracking **One Piece** watch progress without login — equipped with automatic **filler filter**, **saga & arc** navigation map, and a gamified **pirate rank** system.
- [QR Scanner Presence App](#): A modern **attendance solution** leveraging **QR Code** scanning for instant participant check-in, built on a **Next.js + Supabase** stack with **real-time** sync.
- [Malmora \(Reseller & Dropship\)](#): A **reseller & dropship** marketplace tailored for Muslim lifestyle products, enabling anyone to sell without physical stock across **Sulawesi and Eastern Indonesia**.
- [Email Server Makassar Gov](#): A city-scale **Mailcow**-based email cluster serving **200+ official domains** for Makassar City Government, hardened with **SPF/DKIM/DMARC** security protocols and **high-availability** setup.
- [Helpdesk Kota Makassar](#): An **e-government** system for handling **citizen complaints** digitally — featuring **real-time ticket tracking**, automated escalation, and cross-department integration across **Makassar City agencies**.
- [Virtual Museum Makassar](#): An immersive digital museum experience with **360° panoramic** tours, interactive **3D artifact** collections, and **audio-visual** narratives preserving South Sulawesi cultural

heritage.

- **AI-Haram VR**: A high-precision **virtual reality** simulation of **Al-Masjid Al-Haram** and **Al-Masjid An-Nabawi**, developed in **Unity** and optimized for both **VR headsets** and mobile devices as a Hajj & Umrah educational tool.
- **Paralluta (Lalu Lintas)**: A smart platform for **traffic management** in Makassar City — covering **road condition monitoring**, violation logging, and centralized **field officer coordination**.
- **SOP Pelaporan Insiden Siber**: A **SOP** framework and **cyber incident response protocols** designed for **Makassar City Government** IT infrastructure, aligned with **BSSN** and **NIST Cybersecurity Framework** standards.

PROFESSIONAL EXPERIENCE

Regional Office of Ministry of Religious Affairs (South Sulawesi) | Makassar, Indonesia *Data Specialist (Systems & Data)* | 2025 – Present

- Architected the regional data warehouse (Data Sulsel), consolidating religious service data for 9M+ residents into a centralized PostgreSQL repository.
- Developed automated ETL pipelines using Python to sync legacy regional databases with modern cloud-native data platforms, ensuring 100% data consistency.
- Optimized complex query performance and indexing strategies for government reporting systems, reducing reporting latency by over 70%.
- Implemented automated data quality checks and validation protocols for high-stakes public records.

Dinas Kominfo & Disbud Makassar | Makassar, Indonesia *Infrastructure & Data Expert* | 2023 – 2025

- Engineered scalable data storage and ingestion solutions for urban complaint management platforms, handling high-volume multimedia data.
- Managed complex on-premise database environments for 200+ government units, ensuring 99.9% data durability and availability.
- Led data migration projects during critical city-wide infrastructure upgrades, maintaining zero data loss.

GetKasir (Enterprise POS Startup) | Makassar, Indonesia *Co-Founder (Data Architecture)* | 2019 – Present

- Architected the data layer for a 100+ module enterprise retail ecosystem, supporting millions of high-concurrency daily transactions.
- Designed high-performance data replication and sync protocols between thousands of local POS nodes and centralized cloud data warehouses.
- Standardized data governance, schema management, and documentation practices to ensure long-term scalability and reporting accuracy.
- Built a custom analytics engine to provide real-time business intelligence for thousands of retail merchants.

EDUCATION

Bachelor of Computer Science (S.Kom) GPA: 3.22 / 4.00

CERTIFICATIONS

- **National Cyber Exercise #10** - BSSN (2024)
- **Samsung Next Apps Winner** - Samsung Indonesia (2017)
- **Microsoft CityApp Participant** - Microsoft

Education

- **Bachelor of Computer Science (S.Kom)** | UIN Alauddin Makassar *GPA: 3.22 / 4.00*
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Data Engineer Expertise Keywords

Python, SQL, Apache Airflow, dbt, Apache Spark, Kafka, BigQuery, Snowflake, Redshift, ETL/ELT Pipelines, Data Warehousing, Data Modeling, PostgreSQL, GCP, AWS, Docker, Kubernetes, CI/CD for Data, Data Quality, Distributed Computing.

Future-Ready Technical Roadmap

To stay competitive in the global DevOps & Infrastructure market, I am actively implementing:

- **Zero Trust Architecture:** Identity-aware proxies and 'Never Trust, Always Verify' principles.
- **eBPF Security:** Deep kernel-level observability and real-time security monitoring.
- **GitOps:** Using ArgoCD/Flux for automated and declarative infrastructure synchronization.
- **Serverless Security:** Securing AWS Lambda/Edge functions and automated IAM least-privilege auditing.